

BALL SPEED CHECKER

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This circuit measures the speed of a cricket ball based on the time taken by the ball to travel the distance from the bowling crease to the batting crease.

As shown in Fig. 1, the circuit is built around two NE555 timer ICs (IC1 and IC2), 74C926 (IC3), LTS543 (display) and a few discrete components.

The first NE555 timer (IC1) is wired in bistable mode. The output of IC1 is fed to pins 4 and 8 of the second NE555 timer (IC2), which is wired as an astable multivibrator and produces a frequency of 1 kHz. The output of IC2 is counted and displayed by IC3. There are two light barriers built around LDR1 and LDR2, which produce signals when the ball interrupts the laser beam falling on

LDRs while passing from the bowling crease to the batting crease.

When the ball crosses the bowling crease, it momentarily interrupts the IR beam falling on LDR2. This produces a pulse at pin 2 of IC1, which, in turn, sets IC1. The output of IC1 goes high and provides +5V to enable IC2 and it starts oscillating. These pulses are counted by IC3 and shown on 7-segment displays (DIS1 through DIS4).

When the ball crosses the batting crease, it again interrupts the laser beam falling on LDR1. This produces a pulse at pin 6 of IC1 and resets it. The output of IC1 goes low and stops +5V to IC2 to disable it. IC2 stops oscillating, and IC3 also stops counting.

Time is displayed on 7-segment displays. The counter counts from 0 to 9999, i.e., 10 seconds. The distance from the bowling to batting crease is approximately 18 metres. You can now calculate the speed of the bowling ball by substituting the value of distance and time taken by the ball from bowl-

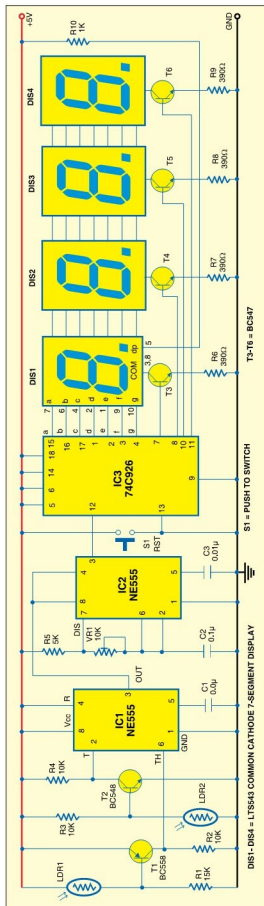


Fig. 1: Speed checker circuit

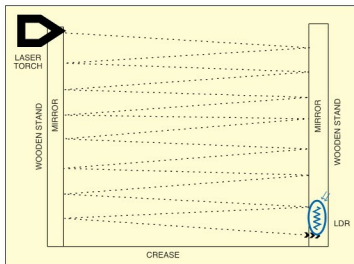


Fig. 2: Sensor fitting at the crease